DiscoverX

# User Manual

# PathHunter®

# Flash Detection Kit

Use with PathHunter Cell Lines or Cell Lines Made with ProLabel®/ProLink™ Vectors

Catalog No. 93-0247 Series

Please read this entire user manual before proceeding with the assay.
For additional information or Technical Support, see contact information below.

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# Overview

The PathHunter Flash Detection Kit is to be used with PathHunter Cell Lines or cell lines made with ProLabel®/ProLink™ expression or cloning vectors. The kit is easy-to-use and has been successfully run in both 96-well and 384-well microplate format. The resulting signal is chemiluminescent and is read with any standard plate reader.

PathHunter products utilize the Enzyme Fragment Complementation (EFC) technology for studying protein-protein interactions, protein translocation, receptor trafficking, and receptor internalization, all of which involve the complementation of two β-galactosidase (β-gal) fragments. The larger β-gal fragment is called Enzyme Acceptor (EA) and the smaller fragment is termed Enzyme Donor (ED; also known as ProLink or ProLabel). The two fragments are inactive when apart. However, when they complement, they form a functional enzyme that hydrolyzes the substrate to generate a chemiluminescent signal. The detection reagents do not contain either the EA or ED fragments of β-gal, and therefore must be used in conjunction with cells that express both components in order to obtain a detectable signal.

# Materials Provided

| **Catalog Number** | **93-0247S** | **93-0247M** | **93-0247** | **93-0247L** | **93-0247XL** |
|---|---|---|---|---|---|
| **Number of Plates** | **1** | **2** | **10** | **100** | **400** |
| Data Points per 96-Well Plate | ~100 | ~200 | ~1,000 | N/A | N/A |
| Data Points per 384-Well Plate | ~400 | ~800 | ~4,000 | ~40,000 | ~160,000 |
| **List of Components** | **Volume per Bottle (mL)** | | | | |
| Flash Cell Assay Buffer | 2.5 | 5 | 25 | 200 | 800 |
| Flash Substrate | 10 | 20 | 100 | 800 | 3200 |
| **Total Volume** | **12.5** | **25** | **125** | **1000** | **4000** |

# Storage Conditions

Upon receipt, store reagents at -20°C. Thaw reagents at room temperature before use. Thawed reagents are stable for 4 days when stored at 2-8°C. The reagents can tolerate up to three freeze-thaw cycles with no impact on performance.

# Additional Materials Required

| **Material** | **Ordering Information** |
|---|---|
| V-Bottom 96-well ligand dilution plates | 92-0011 |
| Multimode or luminescence plate reader | discoverx.com/instrument-compatibility |
| Single and multichannel micro-pipettors and pipette tips | |
| Disposable reagent reservoir | Thermo Fisher Scientific, Cat. No. 8094 or similar |

# Assay Detection Protocol

The following procedure details the assay detection protocol to be used with PathHunter cell lines. Please refer to your cell line user manual for specific cell line and assay preparation details. Detection reagents must be prepared as a working solution prior to use. Once prepared, the working solution is stable for up to 24 hours at room temperature with no impact on assay performance.

| Working Detection Solution (Flash) for 96- or 384-well Format | | |
|---|---|---|
| **Components** | **Volume Ratio** | **Volume per Plate (mL)** |
| Flash Cell Assay Buffer | 1 | 2.5 |
| Flash Substrate | 4 | 10 |
| **Total Volume** | | **12.5** |

1. Prepare a stock of Working Detection Solution (Flash) in a 15 mL polypropylene tube or reagent reservoir by mixing 1-part of Flash Cell Assay Buffer and 4-parts of Flash Substrate.

Make stock within 24 hours of use.

2. Add 110 µL (for 96-well format) or 25 µL (for 384-well format) [equivalent to 100% of the assay volume] of Working Detection Solution (Flash) to all wells of the assay plate.

Do not pipette up and down in the vial to mix or vortex plates.

3. Incubate assay plate for 1 hour at room temperature in the dark.

Working Detection Solution is light sensitive, thus incubation in the dark is necessary.

4. Read samples on a standard luminescence plate reader at 0.1 to 1 second/well for photomultiplier tube readers or 5-10 seconds for imager. The actual signal characteristics over time are affected by lab conditions such as temperature. The user should establish an optimal read time accordingly. Luminescence readout collects signal from all wavelengths. Some instrument manufacturers may include a cutoff filter at high wavelengths, but usually no wavelength setting is needed for luminescence readout.

After addition of reagents, the samples should be read within 2 hours.

5. Data analysis can be performed using your choice of statistical analysis software (e.g. GraphPad Prism, Molecular Devices Softmax Pro, Biotek Instruments Gen5, Microsoft Excel, etc.).

For additional information or technical support, see contact information below.

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